Dirac Delta

$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon A} f\left(\frac{t}{\varepsilon}\right) \quad \text{where} \quad A = \int_{-\infty}^{\infty} f(t)\,dt$$

direc delta = impulse

- $\int_{-\infty}^{\infty} e^{-t^2}\,dt = \sqrt{\pi}$
- $\int_{-\infty}^{\infty} \frac{1}{t^2+1}\,dt = \arctan(t)$
- $\delta(t/a) = \frac{1}{|a|}\delta(t)$
- $f(t)\delta(t) = f(0)\cdot\delta(t)$
- $f(t)\,\delta(t-t_0) = f(t_0)\cdot\delta(t-t_0)$
- $\int_{-\infty}^{\infty} f(t)\,\delta(t-t_0)\,dt = f(t_0)$
- $\frac{du(t)}{dt} = \delta(t)$

- Dot product of two functions: $f(t)\cdot g(t) = \int_{-\infty}^{\infty} f(t)\,g(t)\,dt$ so $\|f(x)\|_{L_2}\cdot\|g(x)\|_{L_2}\cdot\cos\theta$
- Function norm of a function: $\|y\|_{L_p} = \sqrt[p]{\int_a^b |y(x)|^p\,dx}$
- $a+bi = re^{i\theta}$
- $e^{i\theta} = \cos\theta + i\sin\theta$
- $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$
- $\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$

LTI

- $X[n] = \sum X[i]\,\delta[n-i]$
- $\sum x[k]\,\delta[n-k] \rightarrow \boxed{\text{LTI}} \rightarrow \sum x[k]\,h[n-k]$

So $y[0] = \sum x[k]h[-k]$, $y[1] = \sum x[k]h[1-k]$, $y[2] = \sum x[k]h[2-k]$ etc.

$$y[n] = \sum x[k]h[n-k]$$

So reverse h and compute each $y[n]$. See: $h[n] = 2\ \ 1\ \ -1\ \ 2$

$x[n] = 1\ \ 1\ \ \underline{2}\ \ 0\ \ 3$

n:	-2	-1	0	1	2	
$x[n]$:	1	1	$\underline{2}$	0	3	
2	-1	1	$\underline{0}$			$= y[-2] = 0$
	2	-1	1	$\underline{0}$		$= y[-1] = 1$
		2	-1	1	$\underline{0}$	$= y[0] = 0$
			2	-1	1	$= y[1] = \cdots$
				2	-1	
					$\vdots$	

for continuous case we have:

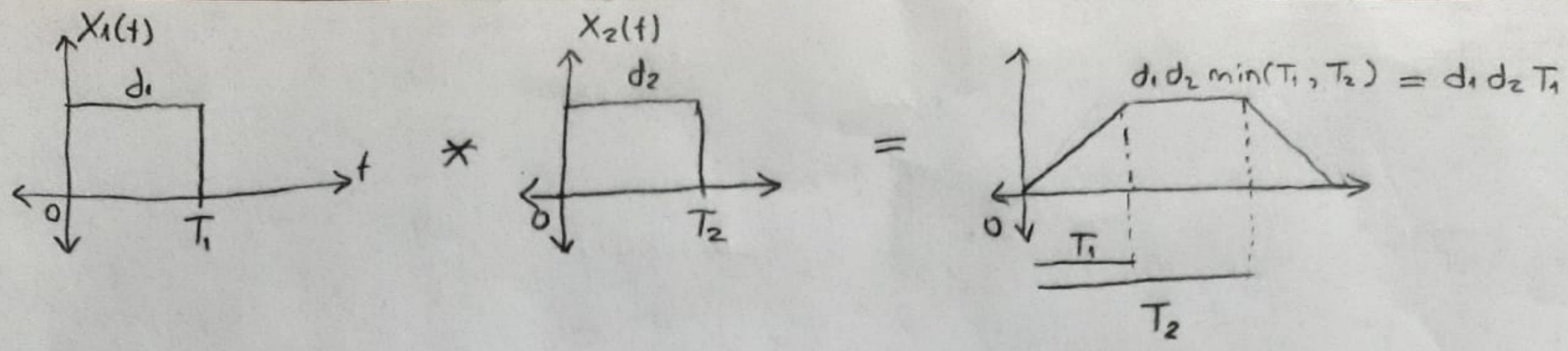
$$y(t) = \int_{-\infty}^{\infty} x(s)\,h(t-s)\,ds \quad \text{or} \quad \int_{-\infty}^{\infty} x(t-s)\,h(s)\,ds \quad \text{where} \quad y(t) = x(t) * h(t)$$

$$\delta(t) \rightarrow \text{LTI (continuous)} \rightarrow h(t)$$

We have also 2D convolution. we reverse both rows and cols of filter.

Size = $n-f+1$

size = $n-f+1+2p$

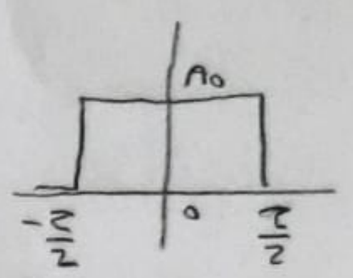


$\int \sin x \, dx = -\cos x$

If $S_n = a + ar + ar^2 + ar^3 + \cdots$ $\quad S_n = \dfrac{a(1-r^n)}{(1-r)}$

Fourier Series - Transform

- revesing t of $X(t)$ will result C_{-n}. (C_n is also reversed)
- $e^{-at}u(t) \xrightarrow{FT} \dfrac{1}{a+\jmath w}$ when $a>0$
- $\lim_{w\to\infty} e^{-iw}$ = NOT DEFINED
- $FT(\text{signum}(t)) = \dfrac{2}{\jmath w}$



$FT = \dfrac{A_0}{w} 2\sin(w\frac{\tau}{2})$

- Fourier transform of a constant A_0 is $2\pi \cdot A_0 \cdot \delta(w)$
- Since $u(t) = (1+\text{signum}(t))/2$ we have $FT(u(t)) = FT(\frac{1}{2}) + FT(\text{signum}(t))/2$

$FT(u(t)) = \pi\delta(w) + \dfrac{1}{\jmath w}$

$w_0 = \dfrac{2\pi}{T}$

($\dfrac{dx(t)}{dt}$) Fourier transform of it eavals $(\jmath w)X(w)$.

- $FT(e^{+\jmath w_0 t}) = 2\pi\delta(w-w_0)$ by using this : $FT(\cos w_0 t) = \pi(\delta(w-w_0) + \delta(w+w_0))$
- $FT(e^{-\jmath w_0 t}) = 2\pi\delta(w+w_0)$ also : $FT(\sin w_0 t) = \jmath\pi(\delta(w+w_0) - \delta(w-w_0))$

- we can find C_n complex exp coefficient of fourier series by using fourier transform. But the function must be a periodic signal.

$C_n = \dfrac{X(nw_0)}{T_0}$ where $X(w_0)$ is the fourier transform of partion T_0, and $X(nw_0)$ is scaled version.

- Fourier transform of periodic continuous signals can be written as: $X(w) = \sum_{n=-\infty}^{\infty} 2\pi C_n \delta(w - nw_0)$ where $C_n = \dfrac{X(nw_0)}{T_0}$.
- Discrete Fourier Series: $X[n] = \sum_{k=<N>} a[k] e^{ik\frac{2\pi}{N}n}$ $\quad a[k] = \frac{1}{N}\sum_{n=<N>} x[n] e^{-ik\frac{2\pi n}{N}}$
- Discrete fourier Transform: $X[\Omega] = \sum_{k=-\infty}^{\infty} x[n] e^{-\jmath w k}$
- Periodic Discrete Fourier Transform $X[\Omega] = 2\pi \sum_{k=-\infty}^{\infty} a_k (\delta(\Omega - \frac{2\pi k}{N}))$
- Inverse continuous fourier transform $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(w) e^{iwt} dw$
- Continuous fourier transform $X(w) = \int_{-\infty}^{\infty} x(t) e^{-iwt} dt$
- Periodic Continuous fourier $X(w) = 2\pi \sum_{k=-\infty}^{\infty} a[k] \delta(w - \frac{2\pi k}{T})$